Laboratory Activity 10
Module 7: Web Security
Title: Secure Login System with Web Security Features (PHP + CSS + JavaScript)

I. OBJECTIVES

At the end of this activity, students should be able to:

1. Understand basic web security concepts
2. Apply authentication and authorization using PHP
3. Use session management for login control
4. Implement client-side validation using JavaScript
5. Improve UI design using CSS
6. Apply basic secure coding practices
7. Understand password hashing concepts

II. SCENARIO

You are tasked to develop a Secure Login System for a small web application.

The system must:

1. Allow users to log in
2. Validate user input
3. Protect against empty or invalid input
4. Secure passwords using hashing
5. Restrict unauthorized access to dashboard

III. REQUIREMENTS

- Login form using HTML
- PHP authentication system
- MySQL database connection
- Password hashing (MD5 concept)
- Session-based login system
- JavaScript validation
- CSS styling for UI design
- Logout functionality

IV. DATABASE SETUP

```
CREATE DATABASE secure_db;

USE secure_db;

CREATE TABLE users (
   id INT AUTO_INCREMENT PRIMARY KEY,
   username VARCHAR(100),
   password VARCHAR(255)
);
```

V. DATABASE CONNECTION (connect.php)

```
<?php
```

```php
// Connect to database

$servername = "localhost";
$username = "root";
$password = "";
$dbname = "secure_db";

$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

VI. CSS FILE (style.css)

```css
/* Page design */
body {
        font-family: Arial;
        background-color: #f4f6f7; t
        ext-align: center;
        padding: 50px;
}
/* Login box design */
.login-box {
        background: white;
        width: 300px;
        margin: auto;
        padding: 20px;
        border-radius: 10px;
        box-shadow: 0px 2px 10px gray;
}
/* Input fields */
input {
        width: 90%;
        padding: 10px;
        margin: 10px 0;
}
/* Button */
button {
        width: 100%;
         padding: 10px;
        background-color: #2c3e50;
        color: white;
         border: none;
}
/* Button hover */
button:hover {
        background-color: #1a252f;
 }
/* Error text */
```

```
.error {
        color: red;
}
```

VII. JAVASCRIPT FILE (script.js)

```
// Validate login form before submitting
function validateLogin() {

    let username = document.getElementById("username").value;
    let password = document.getElementById("password").value;

    // Check empty fields
    if (username.trim() === "" || password.trim() === "") {
        document.getElementById("error").innerHTML =
        "All fields are required!";
        return false;
    }

    // Password length check (security rule)
    if (password.length < 6) {
        document.getElementById("error").innerHTML =
        "Password must be at least 6 characters!";
        return false;
    }

    return true;
}
```

VIII. LOGIN SYSTEM (login.php)

```
<?php
session_start();
include "connect.php";

// Login process
if (isset($_POST['login'])) {

    $username = $_POST['username'];
    $password = $_POST['password'];

    // Check empty input
    if ($username == "" || $password == "") {
        echo "All fields are required!";
    } else {

        // Hash password (basic encryption concept)
        $hashedPassword = md5($password);

        // Check credentials
        $sql = "SELECT * FROM users
              WHERE username='$username'
              AND password='$hashedPassword'";

        $result = $conn->query($sql);
```

```
        if ($result->num_rows > 0) {

            $_SESSION['user'] = $username;

            header("Location: dashboard.php");
        } else {
            echo "Invalid username or password!";
        }
    }
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Secure Login</title>
    <link rel="stylesheet" href="style.css">
    <script src="script.js"></script>
</head>

<body>

<h1>Secure Login System</h1>

<div class="login-box">

<p id="error" class="error"></p>

<form method="POST" onsubmit="return validateLogin()">

    Username:
    <input type="text" id="username" name="username">

    Password:
    <input type="password" id="password" name="password">

    <button type="submit" name="login">Login</button>

</form>

</div>

</body>
</html>
```

IX. DASHBOARD (dashboard.php)

```
<?php
session_start();

if (!isset($_SESSION['user'])) {
    header("Location: login.php");
```

```
    exit();
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Dashboard</title>
</head>

<body>

<h1>Welcome Dashboard</h1>

<?php
echo "Logged in as: " . $_SESSION['user'];
?>

<br><br>

<a href="logout.php">Logout</a>

</body>
</html>
```

X. LOGOUT (logout.php)

```
<?php
session_start();
session_destroy();
header("Location: login.php");
?>
```

XI. CHALLENGE

Students must complete at least 3 tasks:

Security Enhancements

- Add password show/hide toggle
- Improve password rules (uppercase, number required)
- Prevent spaces-only input using .trim()

UI Improvements

- Add background image
- Improve login design (modern UI)
- Add hover animations

FINAL OUTPUT EXPECTATION

- Secure login form
- Input validation (JS + PHP)
- Password hashing
- Session-based access
- Protected dashboard
- Logout system
- Styled UI